

Predicting *E. coli* Drug Resistance Using Traditional Machine Learning and Deep Learning

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Abstract

Antibiotic-resistant *Escherichia coli* (*E. coli*) pose a serious global health threat, as *E. coli* is the most common Gram-negative pathogen in humans and its multidrug-resistant infections are rising worldwide. The increasing prevalence of *E. coli* strains resistant to multiple antibiotics significantly limits treatment options and is associated with higher morbidity and mortality. In this study, we constructed two genomic feature representations: single nucleotide polymorphisms (SNPs) and gene presence/absence (GPA) from a pangenome of 588 *E. coli* genomes. We evaluated five traditional machine learning and one deep learning model for predicting ciprofloxacin resistance across both feature sets. Our results show that models trained on GPA features achieved higher and more consistent performance, while SNP-based models achieved lower accuracy and greater variability, likely due to the high dimensionality and sparsity of SNP data. Additionally, traditional machine learning models consistently outperformed the deep learning approach, suggesting that deep learning methods may require larger datasets to fully capture patterns in genomic data. These findings highlight the importance of feature representation and model selection, and suggest that gene-level features combined with classical machine learning provide an effective and interpretable framework for antimicrobial resistance prediction.

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Chapter 1

Introduction

1.1 Background

Antimicrobial resistance (AMR) is a serious global health problem, making common infections increasingly difficult to treat and driving millions of deaths each year. Recent global reports estimate that more than one million deaths are directly caused by bacterial AMR annually [2]. Improving the speed and accuracy of genomic resistance detection is therefore critical, both to guide appropriate treatment and to support antimicrobial stewardship programs that reduce unnecessary antibiotic use.

Escherichia coli (*E. coli*) is an important organism for studying genomic prediction of AMR. It is one of the most commonly isolated bacteria in clinical laboratories and is responsible for a wide range of infections. In urinary tract infections, *E. coli* is the most frequently reported pathogen [20]. While extraintestinal pathogenic *E. coli* can cause serious

bloodstream infections associated with significant morbidity and mortality [3]. Improving resistance detection in this species therefore has direct clinical relevance.

Ciprofloxacin, a broad-spectrum fluoroquinolone antibiotic, is an important treatment option for serious *E. coli* infections including pyelonephritis and bacteremia [11]. However, resistance to ciprofloxacin in *E. coli* has increased substantially over the past two decades and now exceeds clinically critical thresholds in many regions globally [10, 17]. Resistance arises primarily through chromosomal mutations in the quinolone resistance-determining regions of *gyrA* and *parC* genes, as well as through plasmid-mediated resistance genes such as *qnr* variants [23]. This dual mechanism — chromosomal mutation and horizontal gene acquisition — makes ciprofloxacin resistance in *E. coli* a particularly informative model for comparing genomic feature representations in machine learning.

Traditional antimicrobial susceptibility testing (AST) relies on bacterial culture and phenotypic measurement. Although reliable, this approach is time-consuming and may delay appropriate therapy. Whole-genome sequencing (WGS) offers a faster alternative by providing detailed genomic information, including resistance genes and chromosomal mutations, from which resistance can be predicted directly. Machine learning (ML) methods have shown strong potential for predicting antibiotic resistance from genomic data. Unlike rule-based approaches that rely solely on known resistance genes, ML models can learn complex patterns across genomic variation. A central challenge, however, is how to represent the genome as model input. SNP-based features primarily capture mutation-driven resistance mechanisms, such as chromosomal mutations in *gyrA* and *parC*, whereas gene presence/absence

(GPA) features derived from pangenome analysis reflect horizontally acquired resistance determinants such as *qnr* genes. These two representations capture fundamentally different biological signals, and the choice between them is likely to affect both predictive performance and biological interpretability—factors that are critical for clinical translation.

Although many studies have applied ML to predict antibiotic resistance from genomic data, most evaluate SNP-based and gene presence/absence (GPA) features independently rather than systematically comparing their performance.

1.2 Research Aims

This thesis addresses three specific aims:

1. Construct feature matrices based on SNPs and GPA from a pangenome analysis of 588 *E. coli* genomes.
2. Benchmark five traditional machine learning models and one deep learning model on both feature sets for predicting ciprofloxacin resistance.
3. Evaluate the influence of feature representation and model selection on predictive performance and biological interpretability in genomic AMR prediction.

1.3 Hypothesis

We hypothesise that feature representations based on SNPs and GPA will differ meaningfully in their ability to predict ciprofloxacin resistance in *E. coli*. Given the well-established role of chromosomal mutations in *gyrA* and *parC* as primary drivers of ciprofloxacin resistance, SNP-based features are expected to yield higher predictive performance. GPA features are expected to provide complementary biological interpretability by capturing horizontally acquired resistance determinants. Furthermore, we hypothesise that model performance will vary by feature type, with regularised linear models performing competitively on high-dimensional SNP data and ensemble models better capturing patterns in GPA data.

1.4 Thesis Structure

The remainder of this thesis is organised as follows. Chapter 2 describes the methods, including data sources and dataset construction, pangenome construction and reference genome selection, data preprocessing, machine learning models evaluated, statistical analysis, and experimental setup. Chapter 3 presents the results, covering feature engineering and selection, hyperparameter optimisation, and model performance evaluation. Chapter 4 discusses these findings in the context of existing literature and provides concluding remarks and directions for future work.

Chapter 2

Methods

2.1 Data Sources and Initial Dataset

2.1.1 Origin of Isolates

A total of 600 *Escherichia coli* clinical isolates were obtained from three previously published studies: isolates from invasive *E. coli* in England [13], the NORM surveillance programme in Norway [8], and a US military healthcare network [12]. All isolates were derived from human bloodstream infections (bacteraemia), providing a clinically relevant and consistent infection context across datasets. For each isolate, whole-genome sequencing data and corresponding accession numbers were retrieved from the NCBI and ENA public databases. In addition to genomic sequences, each study provided antimicrobial susceptibility phenotypes determined using the European Committee on Antimicrobial Susceptibility Testing

(EUCAST) clinical breakpoints [7]. This study focused exclusively on ciprofloxacin resistance. Phenotypic resistance labels (Susceptible, Intermediate, Resistant) were as reported in the original studies. As all three studies applied EUCAST-based breakpoints, labels were considered directly comparable across datasets without further harmonisation. Each dataset contained 200 isolates, with ciprofloxacin resistance status used as the binary outcome variable for all downstream analyses.

2.1.1.1 Label Processing and Sample Matching

The initial dataset comprised 600 *E. coli* isolates with phenotype labels including 282 resistant (R), 300 susceptible (S), and 18 intermediate (I) isolates. To simplify the classification task and focus on clinically relevant resistance, intermediate isolates were reclassified as susceptible, following the EUCAST guideline that intermediate isolates may respond to treatment under optimised conditions and are therefore closer in clinical behaviour to susceptible isolates [7]. This resulted in a binary classification task (R vs. S). Prior to feature matrix construction, isolate accession identifiers in the phenotype label file were matched against those in the genomic datasets to ensure consistency. This verification identified 12 isolates that had phenotype labels but no corresponding genomic data, therefore they were excluded from further analysis. After filtering, 588 isolates were retained with both valid genomic data and phenotype labels, comprising 313 susceptible and 275 resistant *E. coli* isolates (Figure 2.1). These matched isolates were used to construct both the SNP and gene presence/absence feature matrices.

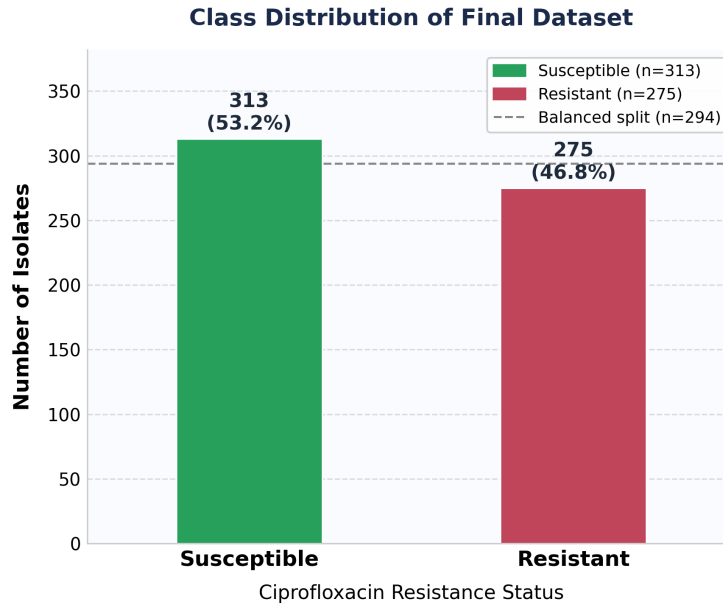


Figure 2.1: Class distribution of the final dataset after label processing and isolate matching. The dataset comprises 313 ciprofloxacin-susceptible and 275 ciprofloxacin-resistant *E. coli* isolates, reflecting a mild class imbalance (53.4% susceptible vs. 46.6% resistant), which motivated the use of Balanced Accuracy as a primary evaluation metric.

2.2 Pangenome Construction and Reference Genome Selection

In collaboration with the CodeLab research team at San Francisco State University, a pangenome was constructed from 588 clinical *E. coli* isolates using Prokka tool for genome annotation and Roary package for pan-genome analysis [18, 15], to characterise gene content variation and identify a representative reference sequence for SNP-based analyses. A pangenome represents the full repertoire of genes observed across a collection of bacterial isolates, encompassing both shared and variable gene content. Gene clusters were categorised

based on their frequency of occurrence across the 588 isolates as follows:

- **Core (99–100%)**: Genes present in nearly all isolates, representing the conserved genomic backbone.
- **Soft-core (95–99%)**: Genes present in most isolates but absent in a small number, likely reflecting minor genomic variation.
- **Shell (15–95%)**: Genes variably present across isolates, potentially reflecting lineage-specific differences or accessory adaptive functions.
- **Cloud (0–15%)**: Rare genes present in only a small subset of isolates, likely representing horizontally acquired or strain-specific elements.

The resulting pangenome comprised 40,575 gene clusters in total, including 1,883 core genes, 367 soft-core genes, 3,842 shell genes, and 34,483 cloud genes (Table 2.1). For this study, we focused on the core, soft-core, and shell gene categories. A reference sequence was constructed by concatenating representative gene sequences from these three categories. Cloud genes were excluded due to their low prevalence across isolates, limited contribution to shared genomic variation, and their tendency to introduce noise in read alignment. This concatenated sequence is referred to hereafter as the *reference shell genome* and was used as the alignment reference for all subsequent SNP-based variant calling.

Table 2.1: Summary of *E. coli* pangenome gene categories and corresponding cluster counts derived from 588 clinical isolates.

Category	Presence (%)	Gene clusters
Core	99–100	1,883
Soft-core	95–99	367
Shell	15–95	3,842
Cloud	0–15	34,483
Total	—	40,575

2.3 Data Preprocessing for SNPs and GPA

We constructed two distinct feature matrices from the shell genome dataset: (1) SNPs matrix representing sequence variations across isolates, and (2) GPA matrix indicating whether each gene is present or absent in each isolate. As illustrated in Figure 2.2, the preprocessing pipeline begins with pangenome construction and proceeds through reference generation and alignment prior to feature extraction.

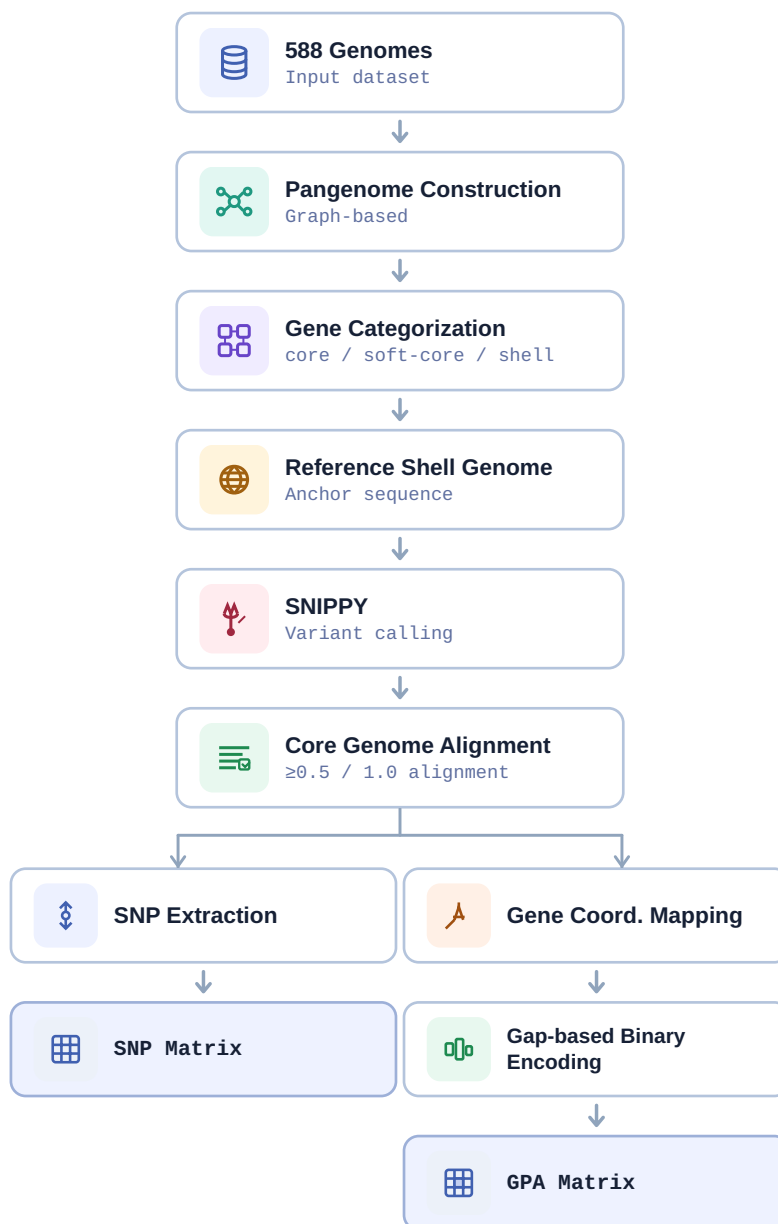


Figure 2.2: Overview of the genomic data preprocessing workflow. Starting from 588 *E. coli* genomes, a pangenome was constructed and categorized into core, soft-core, and shell genes. A reference shell genome was generated and used for variant calling with Snippy, producing a core genome alignment. From this alignment, both the SNP feature matrix and the GPA matrix were derived.

2.3.1 SNP Data Preparation

All 588 isolate genomes were aligned to the reference shell genome using Snippy [19], a reference-based bacterial variant calling pipeline. Snippy generated multiple output files, including a genome alignment `core.full.aln`, `core.ref.fa`, and a tab-delimited SNP summary file `core.tab`. SNP information was extracted from the `core.tab` output file, which summarizes nucleotide states at each core genome position across all isolates. After removing non-genomic metadata columns, the data were transposed to construct an initial SNP matrix of size $588 \times 166,415$, in which rows correspond to isolates and columns correspond to variant sites. Each entry represents the observed nucleotide (A, C, G, T, or gap '-') at a given genomic position. To reduce dimensionality and remove low-quality sites, variant-level filtering was performed using minor allele frequency (MAF) and gap proportion. For each site, MAF was calculated after excluding gap characters as the frequency of the second most frequent non-gap allele. Sites with only one observed non-gap allele were assigned $\text{MAF} = 0$. Gap proportion was defined as the fraction of isolates containing a gap at that position. SNPs were retained if $\text{MAF} \geq 0.05$ and gap proportion ≤ 0.10 . This filtering removed rare variants and poorly aligned sites, resulting in $588 \times 56,717$ high-quality SNPs (a 66% reduction from the original dataset). Each retained SNP was assigned a unique identifier (SNP_0, SNP_1, ...), and genomic coordinates (chromosome, position, reference allele) were stored separately for traceability. For machine learning, nucleotide states were treated as categorical variables and encoded as integer labels prior to model training.

2.3.2 Gene Presence and Absence Data Preparation

The `core.full.aln` file represents a genome alignment (5.7 Mb) in which individual genes from the shell genome are arranged sequentially. To extract gene-specific information from this alignment, we first generated a coordinate mapping for each gene. Using this mapping, we then constructed a binary gene presence/absence (GPA) matrix indicating whether each gene is present or absent in each isolate.

2.3.2.1 Gene Coordinate Mapping

- **Input:** Reference FASTA file (`core_ref.fa`) containing 6,092 gene sequences, each representing a distinct gene cluster in the shell genome.
- **Process:** Each gene i was assigned a start and end coordinate according to:

$$\text{start}_i = \sum_{j=0}^{i-1} \text{length}_j, \quad \text{end}_i = \text{start}_i + \text{length}_i,$$

where length_j is the base-pair length of gene j .

- **Output:** Coordinate mapping table listing `gene_id`, `gene_name`, `start`, `end`, and `length` for all 6,092 genes.

Table 2.2: The gene coordinate mapping table showing start and end positions for the first eight genes.

gene_id	gene_name	start (bp)	end (bp)	length (bp)
FAHFDEJL_03109	<i>rpmG</i>	0	167	168
FAHFDEJL_02987	<i>atpE</i>	168	408	240
FAHFDEJL_00069	<i>gnsA</i>	408	582	174
FAHFDEJL_01249	<i>rpsR</i>	582	810	228
FAHFDEJL_02554	<i>fnr</i>	810	1563	753
FAHFDEJL_02933	<i>atpC</i>	1563	1983	420
FAHFDEJL_00549	<i>hupB</i>	1983	2256	273
FAHFDEJL_03152	<i>fis</i>	2256	2553	297

2.3.2.2 Binary Matrix Generation

Using the gene coordinate mapping, we constructed a binary gene presence and absence matrix in three main steps.

Step 1: Gene region extraction. For each of the 6,092 genes, the corresponding region was extracted from the genome alignment using start and end coordinates. This extraction was performed for all 588 isolates, resulting in a three-dimensional array where each gene's alignment region was represented as a submatrix of size $588 \times L_i$, where L_i is the length of gene i .

Step 2: Gap proportion calculation. For each gene-isolate pair, we calculated the proportion of gap characters (-) in the alignment:

$$\text{gap_proportion}_{ij} = \frac{\text{number of gap characters in gene } i \text{ for isolate } j}{L_i}.$$

Step 3: Binary encoding. A gene was considered *present* if its gap proportion was less than 50% of its length; otherwise, it was marked as *absent*:

$$\text{presence}_{ij} = \begin{cases} 1, & \text{if gap_proportion}_{ij} < 0.5, \\ 0, & \text{otherwise.} \end{cases}$$

The resulting matrix had dimensions $588 \times 6,092$, containing 2,052 core genes (33.7%) and 4,040 accessory genes (66.3%). Feature refinement was performed to reduce redundancy and remove non-informative features prior to machine learning. Genes that showed no variation across isolates (i.e., constant features with ≤ 1 unique value) were removed, as constant inputs cannot contribute to classification. This process removed 2,467 constant genes (40.5%) leaving 3625 features across 588 isolates.

2.4 Machine Learning

2.4.1 Overview of Modeling Approach

In this study, we focused on five supervised machine learning algorithms and a deep learning architecture specifically designed for tabular data to predict ciprofloxacin resistance using two genomic feature matrices: a SNP matrix and a gene presence and absence matrix. The traditional classifiers compared were Decision Tree (DT), Random Forest (RF), Logistic Regression (LR), K-Nearest Neighbors (KNN), and eXtreme Gradient Boosting (XGBoost). These models were selected because they are widely used, interpretable, and well-suited

for classification tasks. Previous genomic antimicrobial resistance studies have used similar algorithms [24]. In addition to these conventional approaches, we implemented TabR, a retrieval-augmented deep learning model for tabular data [9]. TabR augments the prediction process by retrieving relevant training samples (nearest neighbors) and using an attention-like retrieval component to incorporate information from these examples into the final prediction. This mechanism enables the model to capture complex nonlinear patterns and leverages both instance-level similarity and global feature representations, which may be beneficial for high-dimensional genomic feature spaces where traditional models might struggle [9].

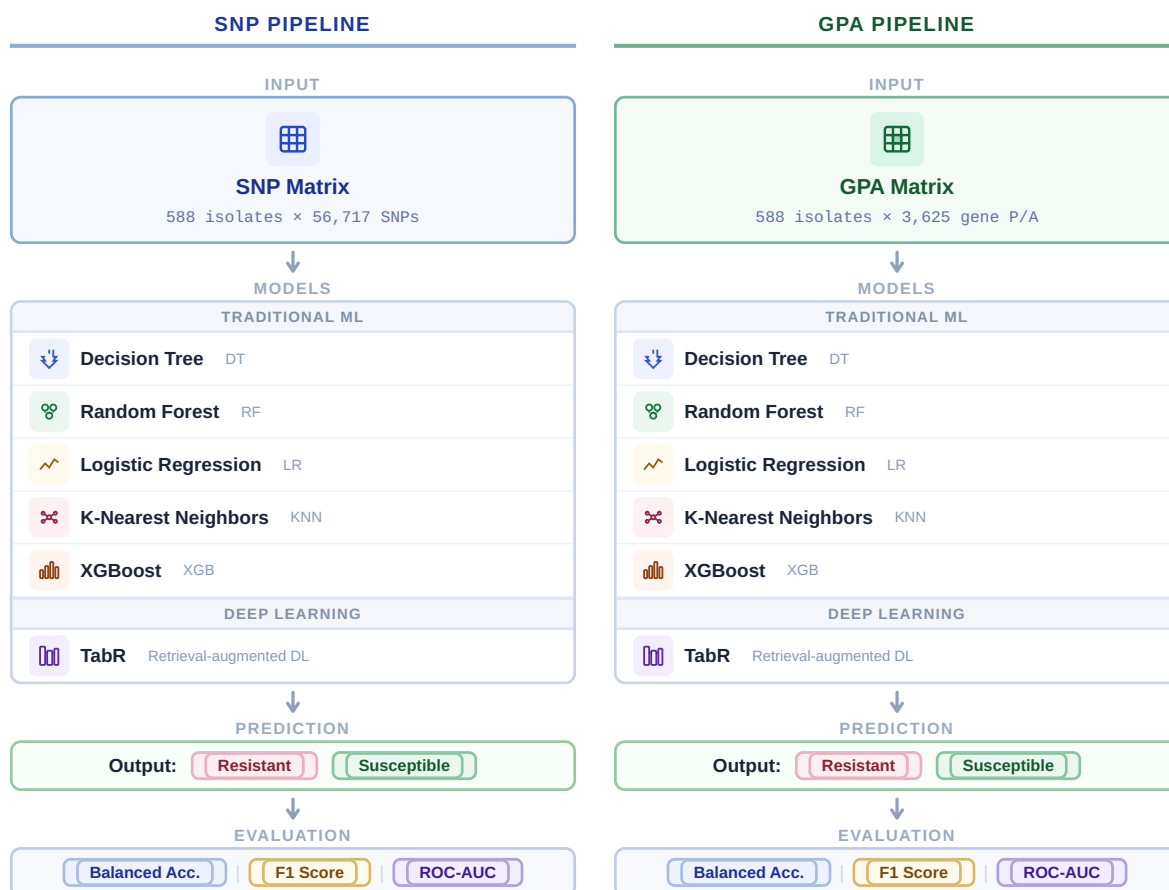


Figure 2.3: Machine learning and deep learning framework for ciprofloxacin resistance prediction. The SNP and GPA matrices serve as inputs to five traditional classifiers and one retrieval-augmented deep learning model (TabR), evaluated using Balanced Accuracy, F1 Score, and ROC-AUC.

By evaluating these models side-by-side across both genomic datasets, we aimed to assess how model choice influences the predictive performance and whether advanced deep learning architectures, like TabR, provide measurable advantages over established machine learning methods in AMR prediction. As illustrated in Figure 2.3, both matrices were evaluated

independently across all six models.

2.4.2 Feature Selection

Feature selection is the process of identifying a subset of relevant features from the original feature space to improve predictive performance and reduce model complexity. In this study, feature selection was applied independently to both the SNP and gene presence/absence datasets. For the SNP dataset, we employed Mutual Information (MI), a filter-based feature selection method that measures the statistical dependency between individual features and the class label [5]. MI scores were computed for each encoded SNP feature independently per cross-validation fold, and features were ranked according to their median MI score across folds. On average, 42,178 SNP features (62.7% of encoded features; range: 41,828–42,679 across folds) were identified as informative above the median threshold. From these, the top 5,000 features per fold were retained for downstream model training, balancing information preservation with computational efficiency. In addition to MI, we considered MultiSURF, a Relief-based feature selection method designed to capture feature interactions. However, MultiSURF was not applied to the SNP dataset due to the computational infeasibility of processing a feature space of this dimensionality.

For the GPA dataset, feature selection was applied using both MI and MultiSURF, computed independently per cross validation fold. MultiSURF was included alongside MI due to its ability to detect feature interaction effects and neighbourhood-based relevance,

which MI alone cannot capture [22]. Features ranked above the median score by either method were retained as informative, yielding on average 2,946 genes (81.3% of the 3,625 filtered genes) for downstream model training, while 678 consistently low-ranked genes were discarded.

2.4.3 Model Selection and Hyperparameter Optimisation

2.4.3.1 Algorithms

Five classical machine learning algorithms and one deep learning architecture were evaluated. The classical models were selected to represent a range of complexity and interpretability, from simple linear models to ensemble methods.

- **Decision Tree (DT)**: A rule-based classifier that recursively partitions the feature space using if-then decisions at each internal node, producing a prediction at each leaf. Included as an interpretable baseline, despite its known susceptibility to overfitting in high-dimensional settings.
- **Random Forest (RF)**: An ensemble of decision trees trained on random feature subsets, aggregating predictions by majority vote. The randomisation reduces variance and improves generalisation over single trees [4].
- **eXtreme Gradient Boosting (XGBoost)**: A sequential boosting ensemble that iteratively fits new models to the residual errors of previous ones, regularised to prevent

overfitting. XGBoost is particularly effective on structured tabular data [6].

- **K-Nearest Neighbors (KNN)**: An instance-based classifier that assigns the majority class among the k nearest neighbours of a query sample in feature space. It requires no explicit training phase but is sensitive to feature scaling and dimensionality.
- **Logistic Regression (LR)**: A linear probabilistic classifier that models resistance probability via a logistic function, providing interpretable feature coefficients and serving as a linear baseline [14].
- **TabR (Retrieval-Augmented Tabular Deep Learning)**: A deep learning architecture designed for tabular data that augments each prediction by retrieving similar training examples via an attention-based mechanism. This allows the model to capture complex non-linear feature interactions that traditional methods may miss, making it well-suited for high-dimensional genomic data [9].

2.4.3.2 Hyperparameter Optimisation

Hyperparameters for all classical ML models were optimised using Optuna, a Bayesian optimisation framework, with the Tree-structured Parzen Estimator (TPE) sampler [1], running 100 trials per algorithm per cross-validation fold. Balanced Accuracy was used as the optimisation objective. All experiments were conducted using a fixed random seed (`random_state = 42`) to ensure reproducibility.

For TabR, each Optuna trial used the fixed training/validation split rather than cross-validation, due to the computational cost of deep learning training. Balanced Accuracy on the validation set was used as the optimisation objective. Tuned hyperparameters included the embedding dimension (`d_main`), context dropout, predictor dropout, learning rate, and weight decay. Final hyperparameter configurations for all models are reported in Appendix A.3.

2.4.4 Model Evaluation

The full labelled dataset ($n=588$) was partitioned into a stratified training set (80%, $n=470$) and a held-out test set (20%, $n=118$), with class proportions preserved in both partitions. The test set was withheld entirely during model development and hyperparameter tuning, and was used solely for final performance evaluation. For classical ML models, 3-fold stratified cross-validation was applied within the training set to assess model stability and guide hyperparameter selection. Each model was subsequently retrained on the full training set using the optimal hyperparameters and evaluated on the held-out test set. For TabR, a fixed train/validation split was used within the training set, with early stopping based on validation loss to prevent overfitting. Final performance was reported on the same held-out test set as the classical models, ensuring direct comparability across all approaches.

Performance was assessed using Balanced Accuracy as the primary metric, which accounts for class imbalance by averaging sensitivity and specificity. We also used additional

metrics including Accuracy, Precision, Recall, F1-Score, and ROC-AUC. Balanced Accuracy was prioritised given the mild class imbalance in the dataset (53.4% susceptible vs. 46.6% resistant).

2.4.5 Model Interpretability

To provide insight into the genomic features driving resistance predictions, model interpretability was assessed through feature importance analysis for all classical ML models across both the SNP and GPA datasets. Feature importance was computed independently per cross-validation fold using model-specific methods:

- **Decision Tree and Random Forest:** Feature importance was derived from the mean decrease in impurity (Gini importance), measuring the total reduction in node impurity contributed by each feature across all trees [4].
- **XGBoost:** Feature importance was computed using the built-in gain metric, which measures the average improvement in model performance when a feature is used for splitting [6].
- **Logistic Regression:** Feature importance was derived from the absolute values of the model coefficients, reflecting the magnitude of each feature's linear association with the resistance label.

- **K-Nearest Neighbors:** Feature importance was approximated using permutation importance, which quantifies the decrease in Balanced Accuracy when each feature’s values are randomly permuted across the test set.

To enable comparison across models with different importance scales, scores were normalized to the range $[0, 1]$ per model per fold, and summarized using the mean rank across folds to account for fold-to-fold variability. A weighted composite importance score was subsequently computed by multiplying each model’s normalized importance by its Balanced Accuracy on the validation set, such that features ranked highly by better-performing models contribute more to the final ranking. The top 20 features based on this weighted composite score were identified for each dataset. Note that interpret-ability analysis was not performed for TabR, as the retrieval-augmented attention mechanism does not produce feature importance scores directly comparable to those of classical ML models.

2.5 Experimental Setup

Datasets. As described in Section 2.3, two datasets were used: (1) a SNP dataset comprising 67,259 encoded features, of which 42,178 were identified as informative after feature selection; and (2) a GPA dataset with 2,946 binary features after feature selection. Each dataset was partitioned into a training set (80%, $n = 470$) and a stratified held-out test set (20%, $n = 118$), with class proportions preserved in both partitions.

Cross-validation. Classical machine learning models were trained using 3-fold stratified cross-validation (CV) on the training set. Hyperparameters were selected to maximize Balanced Accuracy across CV folds. The final performance was evaluated on the held-out test set, which was never used during training or hyperparameter optimization.

Algorithms and optimization. Five classical ML algorithms were compared: Decision Tree (DT), Random Forest (RF), Logistic Regression (LR), XGBoost (XGB), and K-Nearest Neighbors (KNN), alongside the TabR deep learning architecture. Hyperparameters were optimized using Bayesian optimization with the TPE sampler implemented in Optuna [1], with 100 trials conducted per algorithm per CV fold, maximizing Balanced Accuracy.

Evaluation metrics. The primary evaluation metric was Balanced Accuracy. Additional metrics included ROC-AUC, F1 score, Precision, Recall, and Accuracy.

Software and reproducibility. All experiments were conducted using the STREAMLINE automated ML framework [21] with scikit-learn [16], XGBoost [6], Optuna [1], and Python 3.8+. A fixed random seed (`random_state=42`) was used throughout to ensure reproducibility.

Chapter 3

Results

Two genomic feature sets were constructed from 588 matched *E. coli* isolates: a SNP-based matrix (initially 166,415 variants) and a gene presence/absence (GPA) matrix (initially 6,092 genes). Preprocessing reduced dimensionality through quality filtering, one-hot encoding (SNPs only), and feature selection to mitigate overfitting and retain only informative features for ciprofloxacin resistance prediction (Table 3.1).

For the SNP matrix, variants were filtered by minor allele frequency (MAF: 0.05–0.95) and gap proportion ($\leq 10\%$), retaining 56,717 variants, which were subsequently one-hot encoded into 67,259 features. For the gene GPA matrix, genes with missingness exceeding 50% were removed, retaining 3,625 genes. Feature selection was applied independently to each dataset using mutual information (MI) for SNPs and a combination of MI and MultiSURF for GPA features. MultiSURF was included for GPA due to its ability to detect interaction effects between features without requiring a threshold parameter [22]. Features with scores

above the median were retained as informative, yielding a final set of 42,178 SNP features and 2,946 GPA features for model training.

Table 3.1: Summary of preprocessing, feature engineering, and feature selection steps for SNP-based and GPA datasets derived from 588 matched *E. coli* isolates. Filtering thresholds and feature counts after each stage are shown. MI- and MultiSURF-based feature counts are averaged across cross-validation folds.

Statistic	SNP features	Gene P/A features
1. Initial data		
Number of isolates	588	588
Initial features	166,415 variants	6,092 genes
2. Quality filtering		
Minor allele frequency (MAF) threshold	0.05–0.95	—
Gap Proportion	$\leq 10\%$	$< 50\%$
Final features after filtering	56,717 variants	3,625 genes
Features removed (% of initial)	109,698 (65.9%)	2,467 (40.5%)
Feature retention rate	34.1%	59.5%
3. Feature engineering		
One-hot encoded categorical SNPs	67,259 features	—
4. Feature selection		
Method used	MI	MI + MultiSURF
Informative features	42,178	2,946
Uninformative features	25,081	678

3.1 Feature Engineering and Selection

3.1.1 SNP Dataset

After filtering based on minor allele frequency (MAF) and gap proportion (34.1% retention; 56,717 variants), and applying one-hot encoding to multi-allelic loci (i.e., positions

with more than two alleles), a total of 67,259 features were obtained. MI scores identified on average 42,178 informative SNPs (62.7% of encoded features; range: 41,828–42,679 across folds), with 25,081 low-scoring positions discarded. MultiSURF was not applied to the SNP dataset due to the computational infeasibility of applying it to a feature space of this dimensionality. As shown in Figure 3.1, the top-ranked SNP positions exhibited stable MI scores across cross-validation folds, as indicated by the narrow error bars, confirming that feature rankings were consistent and not driven by fold-specific variation. Notably, three SNPs — SNP_17651 (*parC*), SNP_12955_1 (*gyrA*), and SNP_12955_0 (*gyrA*) — showed substantially higher MI scores (0.528–0.549) compared to the remaining top-ranked features ($\Delta \approx 0.25$), suggesting these positions encode the strongest and most consistent association with ciprofloxacin resistance in the dataset. The co-ranking of two features derived from the same *gyrA* position (SNP_12955_1 and SNP_12955_0) reflects one-hot encoding of a multi-allelic variant at this site. These findings are consistent with the well-established role of chromosomal mutations in the quinolone resistance-determining regions of *parC* and *gyrA* as primary drivers of ciprofloxacin resistance in *E. coli* [23]. The remaining top-ranked SNPs showed more modest but relatively uniform scores in the range of 0.234–0.278, mapping to genes including *menB*, *htpG*, *araJ*, and *mgtA*, which may reflect indirect associations or epistatic, but were not explicitly investigated in this study and warrant further investigation.

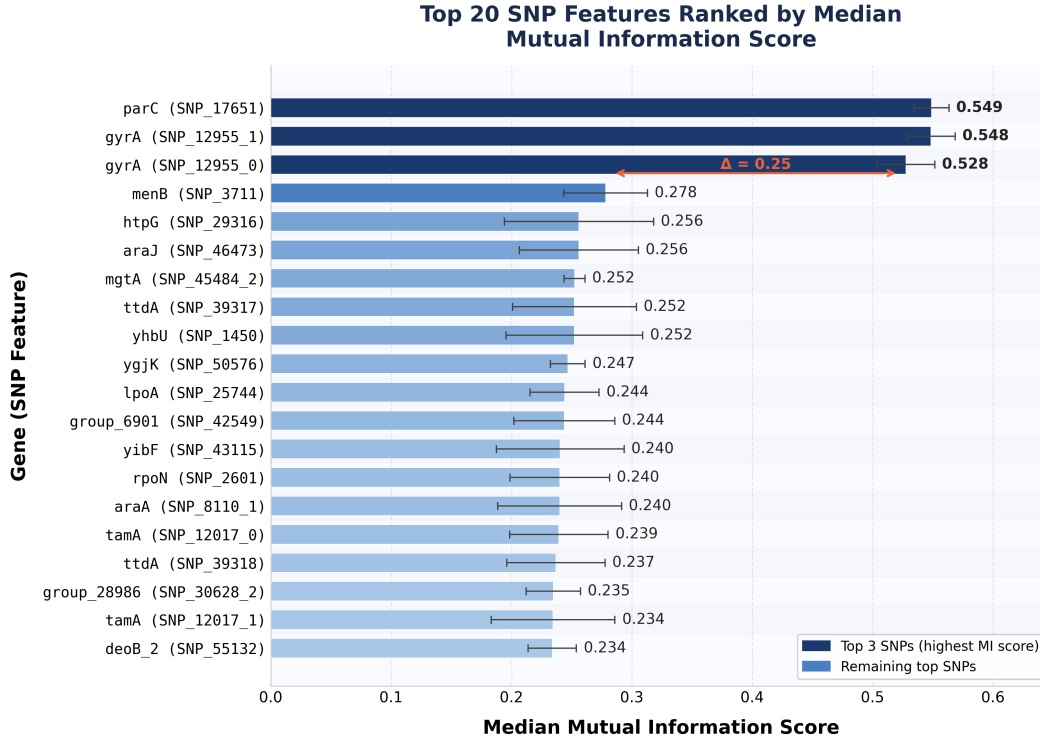


Figure 3.1: Top 20 SNP features ranked by median Mutual Information score across 3-fold cross-validation. Error bars represent the standard deviation of MI scores across folds. The top three SNPs (dark navy) are visually separated from the remaining features by a score gap of $\Delta \approx 0.25$, indicating substantially stronger association with ciprofloxacin resistance. Higher scores indicate greater predictive relevance.

3.1.2 Gene Presence/Absence Dataset

Quality filtering retained 3,625 genes (59.5% of initial features), on which Mutual Information (MI) and MultiSURF scores were computed independently per cross-validation fold. On average, 2,946 genes (81.3% of filtered features) were ranked as informative by both methods using a median score threshold, while 678 genes were consistently low-ranked and

excluded from model training. A summary of all filtering and selection steps is provided in Table 3.1

Figure 3.2 shows the top 20 genes ranked by MI and MultiSURF. Of these 20 genes identified by each method, 13 were shared across both rankings (highlighted in orange), while the remaining genes were method-specific. The overlap within the top-ranked features suggests that MI and MultiSURF identify a partially consistent set of informative genes despite differences in their scoring mechanisms. MI captures marginal statistical dependence between each gene and the resistance label, whereas MultiSURF detects neighborhood-based relevance and feature interactions. The genes identified by both methods — including *argR_2*, *yjgH*, *arcA_2*, *tabA*, *tabA_2*, *bdcA*, *arcC1_3*, *group_26113*, *group_4638*, *group_2537*, *argF*, *argI*, and *group_24131* — represent consistently informative features across both approaches. The method-specific genes may reflect complementary aspects of the genomic signal to which each approach is uniquely sensitive.

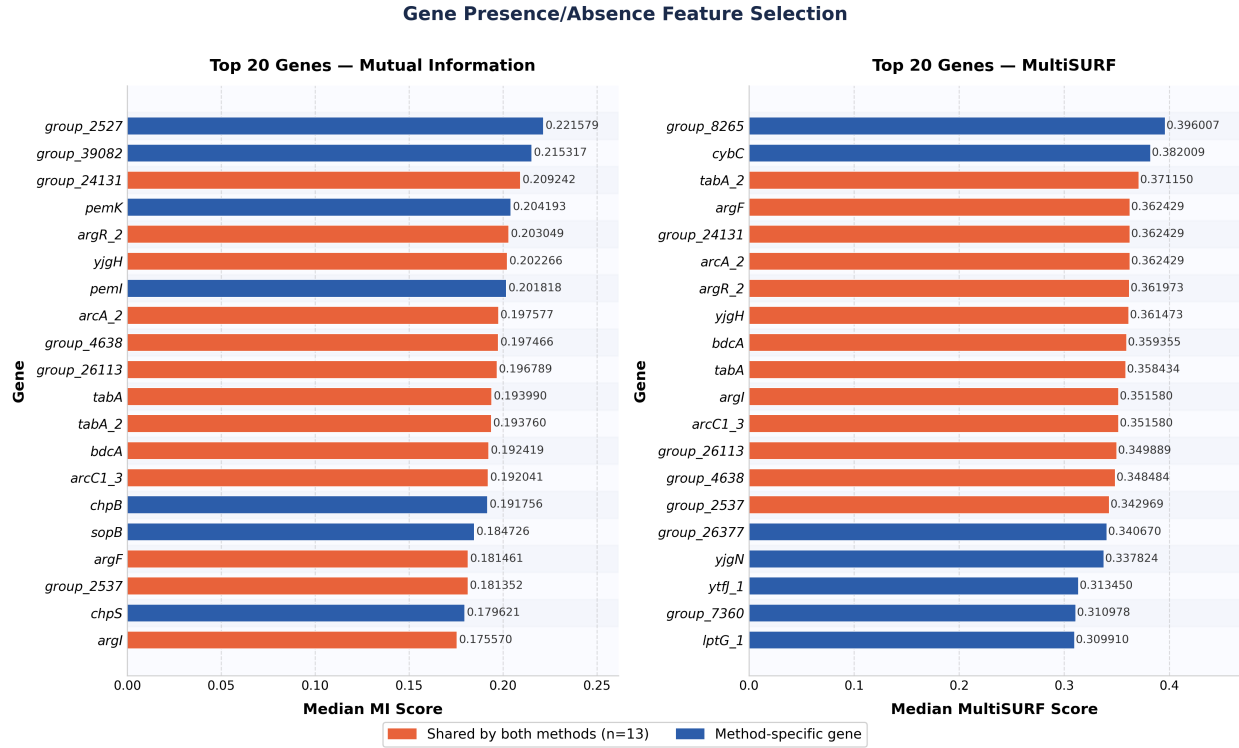


Figure 3.2: Top 20 gene presence/absence features ranked by median Mutual Information (MI) score (left) and median MultiSURF score (right) across cross-validation folds. Genes shared by both methods are highlighted in orange ($n = 13$); method-specific genes are shown in blue. Higher scores indicate greater association with ciprofloxacin resistance.

3.2 Hyperparameter Optimization

We evaluated five classical machine learning models and one deep learning model on both the SNP-based and gene presence/absence datasets. This section focuses on hyperparameter optimization for the classical methods. Five classical models — Decision Tree (DT), K-Nearest Neighbours (KNN), Logistic Regression (LR), Random Forest (RF), and Extreme

Gradient Boosting (XGB) — were tuned via 3-fold stratified cross-validation using Optuna, then retrained on the full training set before test-set evaluation. Each optimization run evaluated balanced accuracy over 100 sampled hyperparameter configurations per model, using a fixed random seed (42) to ensure reproducible folds and search trajectories. For TabR, hyperparameters were configured separately using a fixed architecture and training/validation split, as described in Section 2.4.3.2. Full hyperparameter configurations for each model and dataset are reported in Appendix A.3. Notable observations across models:

- RF and XGBoost hyperparameters were highly stable across folds for the SNP dataset, suggesting a well-defined optimum in that feature space.
- KNN and LR showed greater fold-to-fold variation, particularly for the GPA dataset.
- TabR hyperparameters were identical across both datasets, suggesting architectural robustness to different genomic feature representations.

3.3 Model Performance Evaluation

Model performance was evaluated on an independent held-out test set for both the SNP-based and GPA datasets. All models were trained on the training partition and evaluated on a separate test partition not used during training or hyperparameter tuning, providing an unbiased estimate of generalization to unseen data. The five classical models (DT, KNN, LR, RF, XGB) were retrained on the full training set using their optimized hyperparameters

prior to test-set evaluation. The following subsections present test-set results for each feature representation, followed by a direct comparison across models.

3.3.1 Evaluation of Classical Models on the SNP Dataset

The test set comprised 118 isolates (55 susceptible, class 0, 46.6%; 63 resistant, class 1, 53.4%), with 67,259 one-hot encoded SNP features, representing a near-balanced class distribution. A small number of missing values (470, <0.01% of entries) were imputed during preprocessing.

Table 3.2: Test-set performance on the SNP dataset ($n = 118$). Balanced Acc., balanced accuracy; F1, macro F1 score; ROC-AUC, area under the ROC curve. Highest values per metric are highlighted.

Model	Balanced Acc.	F1 Score	ROC-AUC
Decision Tree	0.916	0.930	0.929
K-Nearest Neighbours	0.843	0.854	0.900
Logistic Regression	0.782	0.827	0.886
Random Forest	0.869	0.879	0.929
XGBoost	0.852	0.849	0.937

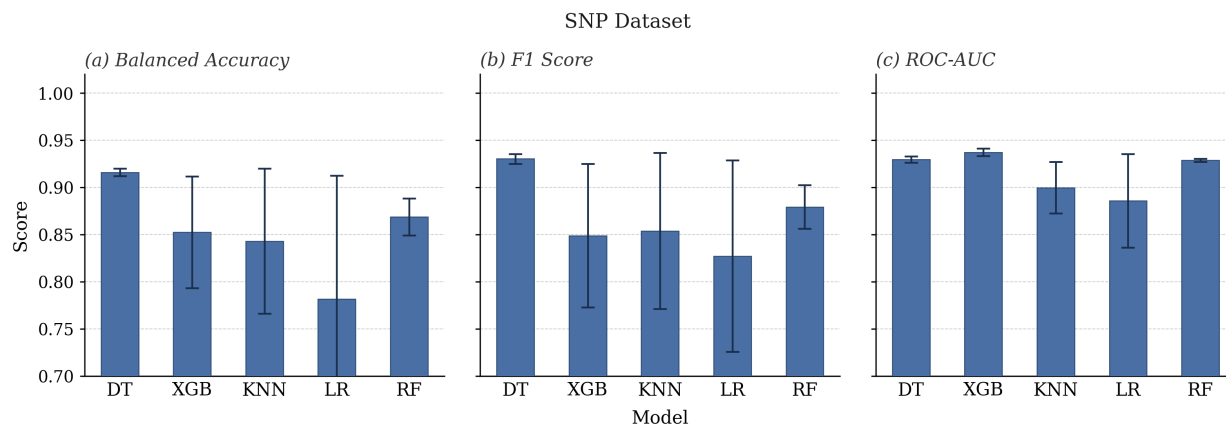


Figure 3.3: Test-set performance of classical models on the SNP dataset across (a) Balanced Accuracy, (b) F1 Score, and (c) ROC-AUC. Error bars show $\pm$ standard deviation across 3-fold cross-validation. DT = Decision Tree, XGB = XGBoost, KNN = K-Nearest Neighbors, LR = Logistic Regression, RF = Random Forest.

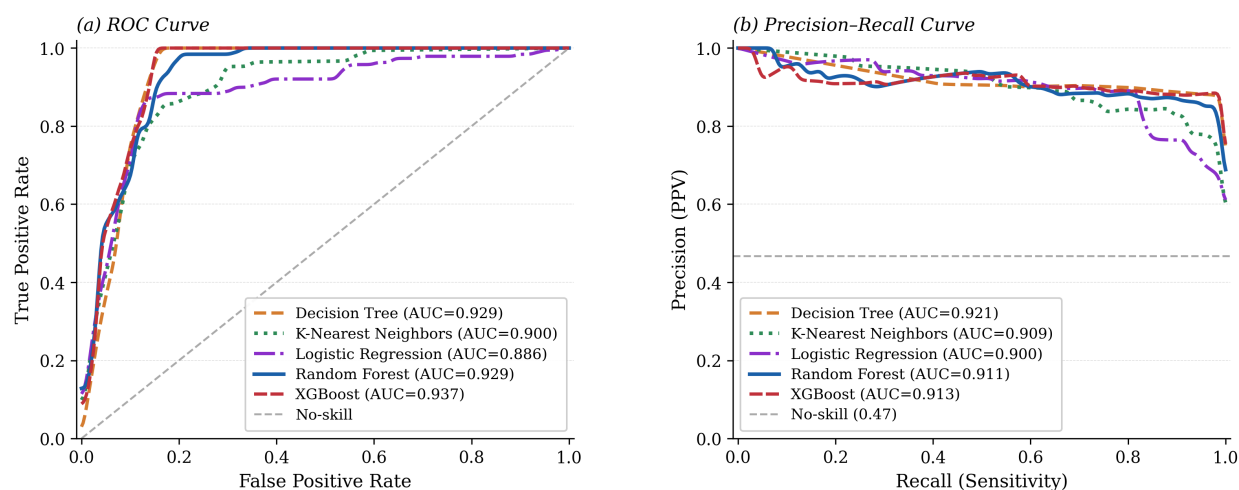


Figure 3.4: ROC and Precision-Recall curves on the SNP test set, averaged across 3-fold cross-validation. (a) ROC curves with area under the curve (AUC) per model. (b) Precision-Recall curves with the no-skill baseline shown as a dashed grey line (prevalence = 0.47). DT = Decision Tree, KNN = K-Nearest Neighbors, LR = Logistic Regression, RF = Random Forest, XGB = XGBoost.

On the SNP test set, Decision Tree achieved the highest balanced accuracy (0.916) and F1-score (0.930) among all classical models (Table 3.2, Figure 3.3), which is a notably different pattern from the GPA dataset where ensemble methods dominated. XGBoost achieved the highest ROC-AUC (0.937), suggesting marginally superior ranking ability across classification thresholds, though its balanced accuracy and F1-score were lower than those of the Decision Tree. Random Forest showed competitive but not superior performance across all metrics. Logistic Regression underperformed across all metrics, likely reflecting the difficulty of learning linear decision boundaries in an extremely high-dimensional categorical feature space. The ROC and Precision–Recall curves (Figure 3.4) are consistent with these findings. Overall, the contrast with the GPA results suggests that localized, non-linear decision rules may better capture discriminative SNP patterns than either linear or ensemble approaches in this setting.

3.3.2 Evaluation of Classical Models on the GPA Dataset

The test set comprised 118 isolates (68 resistant, class 0, 57.6%; 50 susceptible, class 1, 42.4%), with 3,082 binary GPA features, reflecting a moderate class imbalance.

Table 3.3: Test-set performance on the GPA dataset. Highest values per metric are highlighted.

Model	Balanced Acc.	F1 Score	ROC-AUC
Decision Tree	0.876	0.858	0.928
K-Nearest Neighbours	0.929	0.922	0.987
Logistic Regression	0.940	0.933	0.981
Random Forest	0.943	0.937	0.987
Extreme Gradient Boosting	0.934	0.929	0.989

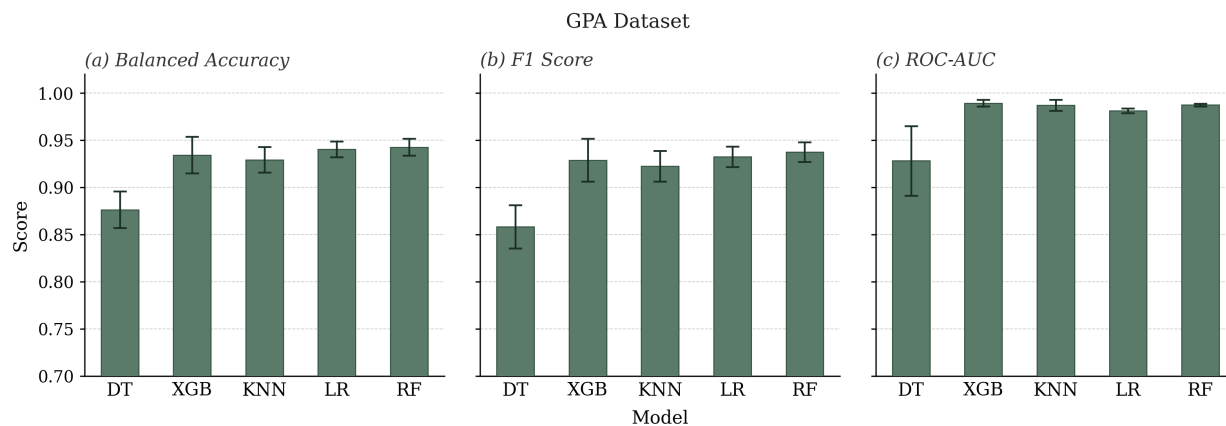


Figure 3.5: Test-set performance of classical models on the Gene Presence/Absence (GPA) dataset across (a) Balanced Accuracy, (b) F1 Score, and (c) ROC-AUC. Error bars show $\pm$ standard deviation across 3-fold cross-validation. DT = Decision Tree, XGB = XGBoost, KNN = K-Nearest Neighbors, LR = Logistic Regression, RF = Random Forest.

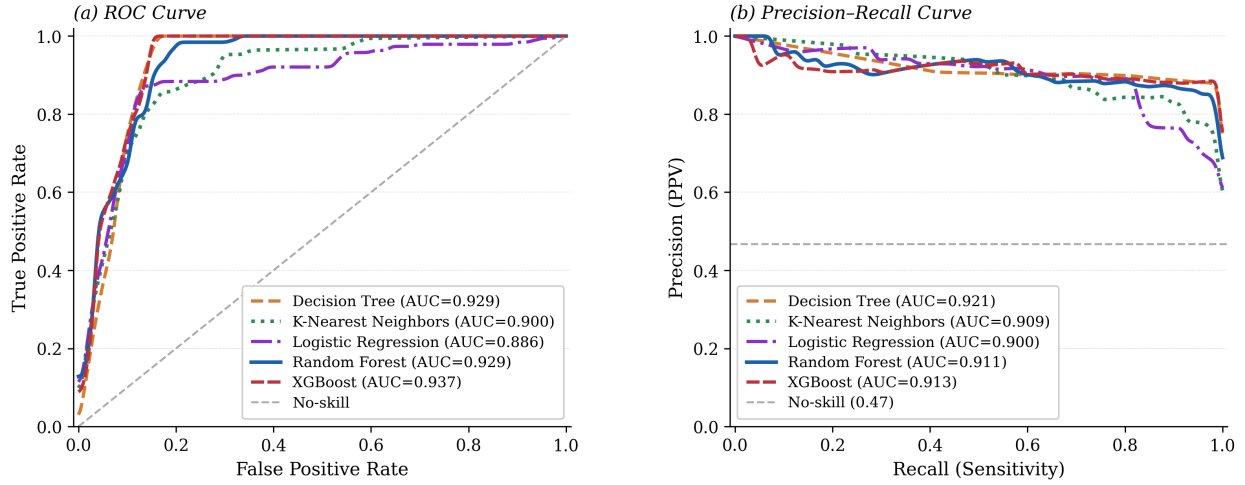


Figure 3.6: ROC and Precision–Recall curves on the GPA test set, averaged across 3-fold cross-validation. (a) ROC curves with area under the curve (AUC) per model. (b) Precision–Recall curves with the no-skill baseline shown as a dashed grey line (prevalence = 0.47). DT = Decision Tree, KNN = K-Nearest Neighbors, LR = Logistic Regression, RF = Random Forest, XGB = XGBoost.

Ensemble-based models demonstrated superior predictive performance on the GPA test set (Table 3.3, Figure 3.5). Random Forest achieved the highest balanced accuracy (0.943) and F1-score (0.937), indicating strong classification across both classes. Extreme Gradient Boosting achieved the highest ROC-AUC (0.989), reflecting slightly superior ranking ability across thresholds. Logistic Regression performed competitively, closely approaching ensemble methods, while K-Nearest Neighbours showed moderate performance. Decision Tree consistently underperformed across all metrics with greater variability across folds. The ROC and Precision–Recall curves (Figure 3.6) confirm these findings, with ensemble methods showing stronger separation from the no-skill baseline and more stable precision at

higher recall levels.

3.3.3 Comparison of Best Classical Model and TabR

The best-performing classical model on each dataset was compared against TabR to assess whether deep learning offers a measurable advantage over traditional methods in this setting. Balanced accuracy and macro F1 score were used as primary metrics, with ROC-AUC reported additionally to evaluate discriminative performance independent of threshold selection.

3.3.3.1 SNP Dataset

On the SNP dataset, Decision Tree was the best classical model, achieving a balanced accuracy of 0.916, macro F1 of 0.930, and ROC-AUC of 0.929 (Table 3.2). TabR achieved a balanced accuracy of 0.872, macro F1 of 0.872, and ROC-AUC of 0.926 (Figure 3.7). The performance gap was more pronounced than on the GPA dataset, with differences of 0.044 in balanced accuracy and 0.058 in macro F1 favouring the classical model. The near-identical ROC-AUC values (0.929 vs. 0.926) suggest that TabR retains comparable ranking ability but struggles with balanced class prediction in the high-dimensional SNP feature space.

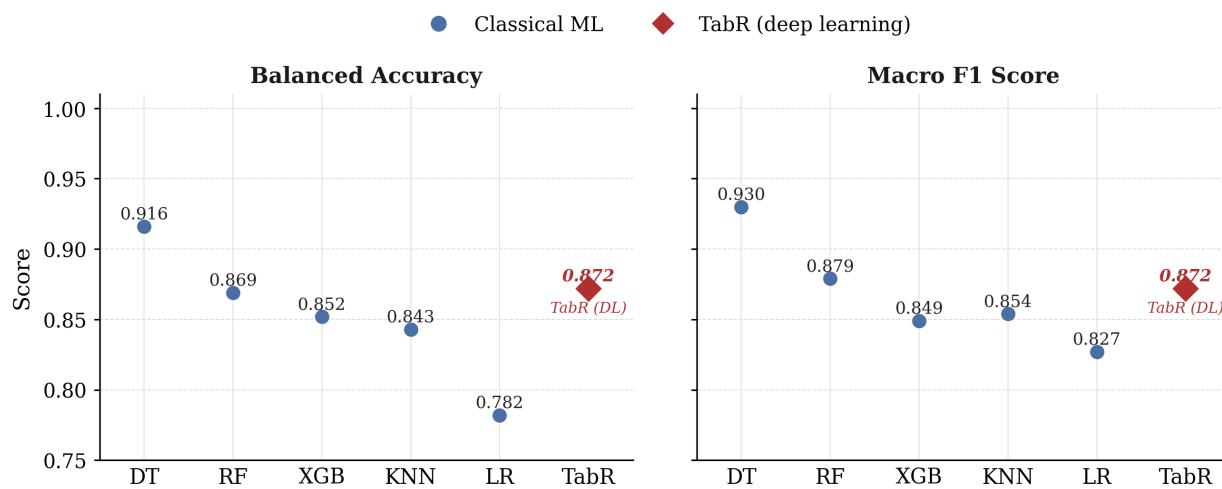


Figure 3.7: Performance comparison on the SNP dataset between classical models (blue) and TabR (red) on the independent test set. Value labels indicate test-set scores.

3.3.3.2 Gene Presence/Absence (GPA) Dataset

On the GPA dataset, Random Forest was the best classical model, achieving a balanced accuracy of 0.943, macro F1 of 0.937, and ROC-AUC of 0.987 (Table 3.3). TabR achieved a balanced accuracy of 0.925, macro F1 of 0.930, and ROC-AUC of 0.974 (Figure 3.8). Random Forest outperformed TabR by 0.018 in balanced accuracy and 0.007 in macro F1, while the ROC-AUC difference was modest (0.013). These results indicate that ensemble tree-based methods retain a small but consistent advantage over deep learning on the GPA representation under the current experimental conditions.

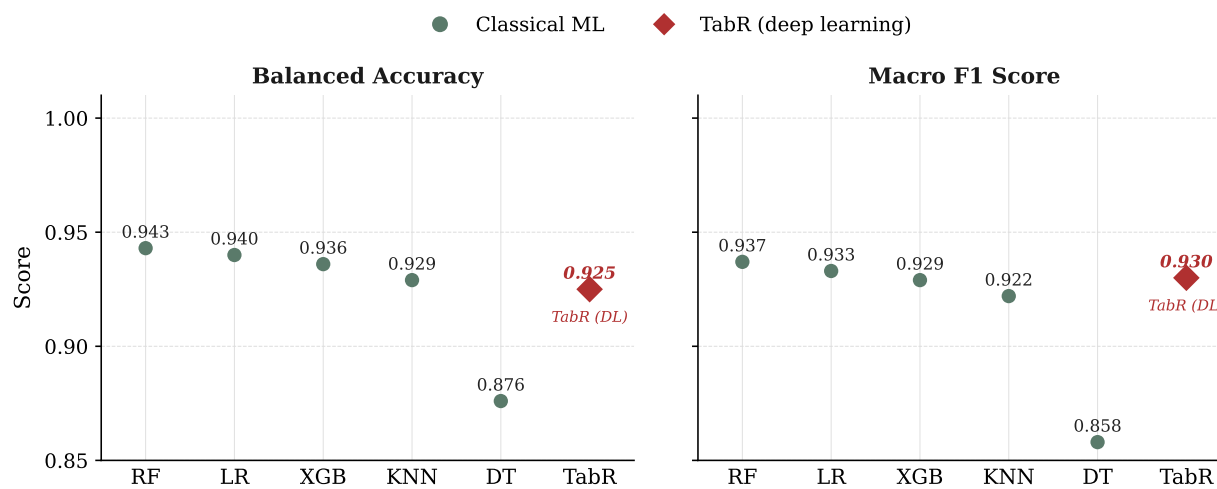


Figure 3.8: Performance comparison on the GPA dataset between classical models (green) and TabR (red) on the independent test set. Value labels indicate test-set scores.

3.3.3.3 Overall Comparison

Across both datasets, classical models outperformed TabR on threshold-dependent metrics, while ROC-AUC differences were consistently small. The performance gap was larger on the SNP dataset, where Decision Tree showed a notable advantage despite the high dimensionality — a pattern not observed on the GPA dataset where ensemble methods dominated. These findings suggest that, for moderate-sized tabular genomic datasets, well-tuned classical models remain competitive with and can outperform deep learning architectures, particularly when feature spaces are large and sparse.

Chapter 4

Conclusion and Future Work

4.1 Conclusion

We investigated two genomic feature representations, SNP and GPA features, and compared their effects on the predictive performance of five traditional machine learning models and one deep learning method for predicting ciprofloxacin resistance in *E. coli*. Our results revealed a clear difference in performance between the GPA and SNP feature representations. Models trained on the GPA dataset achieved higher and more consistent performance across all evaluation metrics (see Appendix A, Figures A.1 and A.2). In contrast, SNP-based models, showed lower overall performance and greater variability across methods. One likely explanation is the complexity of the feature space. The GPA dataset, with a smaller and more structured set of features, may provide clearer signals for classification. In contrast, the SNP dataset has substantially higher dimensionality and is more sparse - a characteristic

that itself reflects the nature of mutation-based genomic data - making it more challenging for most models to learn stable decision boundaries.

Classical machine learning models consistently outperformed the deep learning approach (TabR) on both datasets (Figures 3.7 and 3.8), with a more pronounced performance gap observed in the SNP dataset. This suggests that, under the current dataset size and feature characteristics, tree-based models are better suited for handling high-dimensional genomic data, while deep learning models may require larger datasets or, more structured representations to perform optimally. Interestingly, the relatively small differences in ROC-AUC models indicate that most approaches were able to similarly rank resistant and susceptible isolates. However, the larger gaps observed in balanced accuracy and macro F1 score suggest that Logistic Regression and K-Nearest Neighbors in particular struggled with threshold-dependent classification in the SNP features.

The feature importance analysis provides further biological insight into the differences between the two representations. In the SNP dataset, the highest-ranked features were strongly concentrated in well-known resistance-associated genes, particularly *gyrA* and *parC* (Figure 3.1). These SNPs showed substantially higher importance scores than other features, indicating that a small number of chromosomal mutations carry a strong and consistent signal for ciprofloxacin resistance. This observation aligns with the established role of mutations in quinolone resistance-determining regions as primary drivers of resistance in *E. coli*. In the GPA dataset, by contrast, a broader set of informative features was distributed across multiple genes. A substantial overlap between Mutual Information and MultiSURF rankings

further suggests that these methods converge on a core set of consistently informative genes (Figure 3.2). However, unlike the SNP results, no single gene or small group of genes exhibited a similarly strong effect in the GPA dataset. This pattern indicates that GPA features may capture more distributed or indirect signals, potentially reflecting accessory genome variation or secondary biological processes associated with resistance.

The performance achieved in this study is broadly consistent with, and in some respects exceeds, prior work on genomic AMR prediction in *E. coli*. Mordaigaravand *et al.* [13] applied four ML models — logistic regression, random forests, gradient boosted decision trees, and deep neural networks — to 1,936 *E. coli* isolates phenotyped for resistance to 11 antibiotics, using GPA, population structured, SNP and year of isolation as features. For ciprofloxacin specifically, their best model achieved an accuracy of 0.93, a susceptibility F1 score of 0.95, and a resistance F1 score of 0.84 [13]. Notably, they found that adding SNP data improved ciprofloxacin prediction accuracy by 2.8%, with variants in *gyrA* and *parC* identified as the most important features — a finding that aligns closely with our SNP feature importance results. Consistent with their observations, our study also found that deep learning did not substantially outperform classical ML models, and that gene-level features provided strong predictive signal for ciprofloxacin resistance. Our GPA-based models achieved a balanced accuracy of up to 0.943 and a ROC-AUC of 0.987 on a dataset of 588 isolates, suggesting that GPA features derived from a pangenome reference can yield competitive performance even with a reduced sample size. Similarly, Orcales *et al.* [14] demonstrated that GPA combined with classical ML models provides an effective and interpretable framework

for AMR prediction in *E. coli*, further supporting the utility of this feature representation. Taken together, these comparisons suggest that our approach is methodologically consistent with the current work, while contributing a direct evaluation of SNP and GPA feature representations under a unified experimental framework.

4.2 Limitations

Despite the promising results, this study has several for improvement. First, the original study design intended to construct a complete pangenome from all available *E. coli* isolates; however, computational constraints (e.g., memory requirements and runtime limitations for large-scale pangenome construction) and data quality issues (e.g., incomplete assemblies, missing gene annotations, and variability in sequencing quality) required filtering and quality control steps that reduced the final dataset to 588 isolates. This reduction in sample size may have affected the diversity of the pangenome and the representativeness of the resulting SNP and GPA feature matrices. Related to this, a complete species-wide pangenome of *E. coli* was not constructed. Due to the high genomic diversity of *E. coli* and the computational complexity associated with large-scale pangenome construction, the analysis was instead focused on the shell genome derived from the available isolates. While this approach enabled efficient feature extraction and model training, it may not fully capture the entire spectrum of genetic variation present across all *E. coli* strains.

Second, SNP and GPA features were evaluated independently, and potential interactions

between these feature types were not explored. Integrating both representations may capture complementary biological signals and improve predictive performance. Third, the dataset, while sufficient for training classical ML models, comprises 588 isolates, which may limit the performance of deep learning approaches such as TabR that typically require larger datasets to generalize effectively. Feature selection was also constrained by computational considerations, as MultiSURF could not be applied to the SNP dataset due to its high dimensionality, potentially limiting the ability to capture feature interactions. Furthermore, the use of a median-based threshold for feature selection may have excluded lower-ranked features that still carry predictive information. Finally, rare genes in the GPA dataset may introduce sparsity and reduce the reliability of certain features.

4.3 Future Work

Several directions could build on the findings of this study. First, future work could explore scalable pangenome construction methods or distributed computing approaches to incorporate a more comprehensive species-wide genomic representation of *E. coli*, potentially improving both feature diversity and model generalizability beyond what was achievable with the current 588-isolate dataset. Second, integrating SNP and GPA features into a unified representation may capture complementary biological signals and improve predictive performance beyond what either representation achieves independently. Third, applying more advanced deep learning architectures — such as transformer-based models or graph neural

networks designed for genomic data — may better exploit the high-dimensional structure of SNP data and overcome the limitations observed with TabR under the current dataset size. Fourth, expanding the dataset by incorporating additional publicly available *E. coli* bloodstream infection isolates would provide a more robust evaluation of deep learning approaches and improve the generalizability of the trained models. Finally, extending the framework to predict resistance to multiple antibiotics simultaneously, rather than focusing exclusively on ciprofloxacin, would broaden the clinical utility of the approach.

Bibliography

- [1] Takuya Akiba, Shotaro Sano, Toshihiro Yanase, Takeru Ohta, and Masanori Koyama. Optuna: A next-generation hyperparameter optimization framework. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 2623–2631, 2019.
- [2] Antimicrobial Resistance Collaborators. Global burden of bacterial antimicrobial resistance in 2019: a systematic analysis. *The Lancet*, 399(10325):629–655, January 2022.
- [3] Marc Bonten, James R. Johnson, Anita H. J. van den Biggelaar, Leonidas Georgalis, Jeroen Geurtsen, Patricia Ibarra de Palacios, Stefan Gravenstein, Thomas Verstraeten, Peter Hermans, and Jan T. Poolman. Epidemiology of *escherichia coli* bacteremia: A systematic literature review. *Clinical Infectious Diseases*, 72(7):1211–1219, 2021.
- [4] Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.
- [5] Gavin Brown, Adam Pocock, Ming-Jie Zhao, and Mikel Luján. Conditional likelihood maximisation: A unifying framework for information theoretic feature selection. *Journal*

of Machine Learning Research, 13:27–66, 2012.

- [6] Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, San Francisco, CA, USA, August 2016.
- [7] EUCAST. Redefining susceptibility testing categories S, I and R, version 9.0. <https://www.eucast.org/bacteria/clinical-breakpoints-and-interpretation/definition-of-s-i-and-r/>, 2019. Accessed: 2026.
- [8] Rebecca A. Gladstone, Alan McNally, Anna K. Pöntinen, Gerry Tonkin-Hill, John A. Lees, Kusti Skytén, François Cléon, Martin O. K. Christensen, Bjørg C. Haldorsen, Kristina K. Bye, Karianne W. Gammelsrud, Reidar Hjetland, Angela Kümmel, Hege E. Larsen, Paul Christoffer Lindemann, Iren H. Löhr, Åshild Marvik, Einar Nilsen, Marie T. Noer, Gunnar S. Simonsen, Martin Steinbakk, Ståle Tofteland, Marit Vattøy, Stephen D. Bentley, Nicholas J. Croucher, Julian Parkhill, Pål J. Johnsen, Ørjan Samuelsen, and Jukka Corander. Emergence and dissemination of antimicrobial resistance in escherichia coli causing bloodstream infections in norway in 2002–17: a nationwide, longitudinal, microbial population genomic study. *The Lancet Microbe*, 2(7):e331–e341, 2021.
- [9] Yury Gorishniy, Ivan Rubachev, Nikolay Kartashev, Daniil Shlenskii, Akim Kotelnikov, and Artem Babenko. Tabr: Unlocking the power of retrieval-augmented tabular deep

- learning. *arXiv preprint arXiv:2307.14338*, 2023.
- [10] Amira M. Kamel et al. Ciprofloxacin resistance among uropathogenic *Escherichia coli* isolates. *BMC Infectious Diseases*, 21:996, 2021.
- [11] Barbara Kot. Antibiotic resistance among uropathogenic *Escherichia coli*. *Polish Journal of Microbiology*, 68(4):403–415, 2019.
- [12] Emma G. Mills, Melissa J. Martin, Ting L. Luo, Ana C. Ong, Rosslyn Maybank, Brendan W. Corey, Casey Harless, Lan N. Preston, Joshua A. Rosado-Mendez, Scott B. Preston, Yoon I. Kwak, Michael G. Backlund, Jason W. Bennett, Patrick T. McGann, and François Lebreton. A one-year genomic investigation of *Escherichia coli* epidemiology and nosocomial spread at a large us healthcare network. *Genome Medicine*, 14(1):147, 2022.
- [13] Danesh Moradigaravand, Martin Palm, Anne Farewell, Ville Mustonen, Jonas Warringer, and Leopold Parts. Prediction of antibiotic resistance in *Escherichia coli* from large-scale pan-genome data. *PLOS Computational Biology*, 14(12):e1006258, 2018.
- [14] Faye Orcales, Lucy Moctezuma Tan, Meris Johnson-Hagler, John Matthew Suntay, Jameel Ali, Kristiene Recto, Phelan Glenn, and Pleuni S. Pennings. Using genomic data and machine learning to predict antibiotic resistance: A tutorial paper. *PLOS Computational Biology*, 20(12):e1012579, 2024.

- [15] Andrew J Page et al. Roary: rapid large-scale prokaryote pan genome analysis. *Bioinformatics*, 31(22):3691–3693, 2015.
- [16] Fabian Pedregosa et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- [17] Wieneke C. Rottier et al. *Escherichia coli* resistance to fluoroquinolones in community-acquired urinary tract infection. *Antimicrobial Agents and Chemotherapy*, 65(1), 2021.
- [18] Torsten Seemann. Prokka: rapid prokaryotic genome annotation. *Bioinformatics*, 30(14):2068–2069, 2014.
- [19] Torsten Seemann. Snippy: rapid haploid variant calling and core genome alignment. GitHub repository, 2015. accessed 2026-02-15.
- [20] Morgan R. Timm, Seongmi K. Russell, and Scott J. Hultgren. Urinary tract infections: pathogenesis, host susceptibility and emerging therapeutics. *Nature Reviews Microbiology*, 23(2):72–86, 2025.
- [21] Ryan J. Urbanowicz et al. STREAMLINE: A simple, transparent, end-to-end automated machine learning pipeline. *arXiv preprint arXiv:2206.12002*, 2022.
- [22] Ryan J. Urbanowicz, Randal S. Olson, Peter Schmitt, Melissa Meeker, and Jason H. Moore. Benchmarking relief-based feature selection methods for bioinformatics data mining. *Journal of Biomedical Informatics*, 85:168–188, 2018.

- [23] Ann Versporten et al. Quantifying the contribution of resistance mechanisms to ciprofloxacin MIC in *Escherichia coli*. *Journal of Antimicrobial Chemotherapy*, 74(2):298–310, 2019.

- [24] Yi Xu, Ying Mao, Xiaoting Hua, Yan Jiang, Yi Zou, Zhichao Wang, Zubi Liu, Hongrui Zhang, Lingling Lu, and Yunsong Yu. Machine learning-based prediction of antimicrobial resistance and identification of amr-related snps in *Mycobacterium tuberculosis*. *BMC Genomic Data*, 26(1):48, 2025.

Appendix A

Additional Results

A.1 SNP Dataset — Model Performance Heatmap

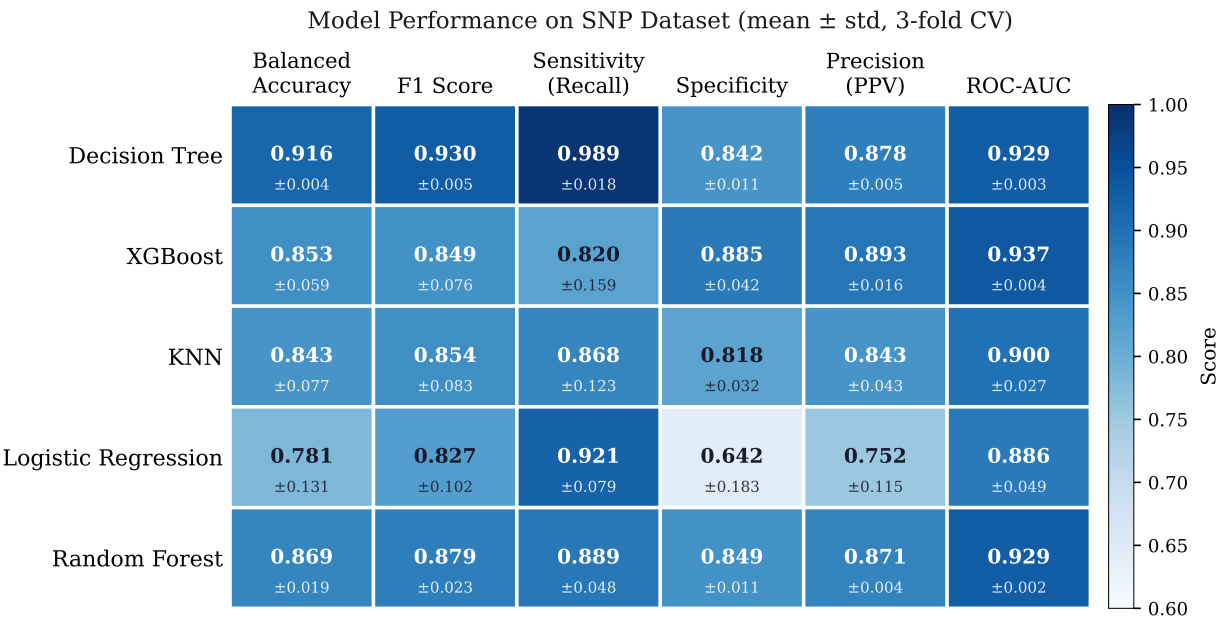


Figure A.1: Model performance heatmap on the SNP dataset. Each cell reports mean $\pm$ standard deviation across 3-fold cross-validation. Darker shading indicates higher performance.

A.2 GPA Dataset - ML Model Performance Heatmap

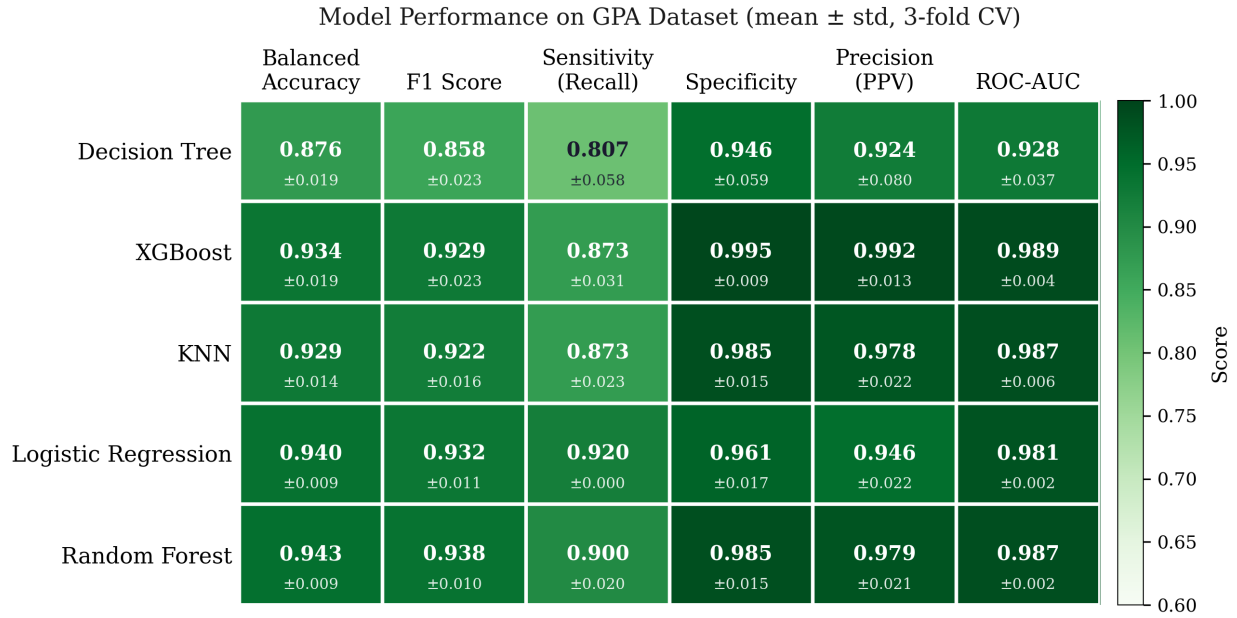


Figure A.2: Model performance heatmap on the Gene Presence/Absence (GPA) dataset. Each cell reports mean $\pm$ standard deviation across 3-fold cross-validation. Darker shading indicates higher performance.

A.3 Selected Hyperparameters

Table A.1 and Table A.2 report the best hyperparameters identified by Optuna for each model across both datasets.

Table A.1: Best hyperparameters selected by Optuna for each classical ML model across 3-fold cross-validation folds (F0, F1, F2) on the SNP and gene presence/absence (GPA) datasets. Dashes (–) indicate that the parameter was not applicable or not selected for that fold.

Model	Hyperparameter	SNP Dataset			Gene P/A Dataset		
		F0	F1	F2	F0	F1	F2
DT	criterion	gini	gini	gini	gini	entropy	entropy
	max_depth	29	4	29	12	21	19
	min_samples_split	45	26	45	13	23	11
	min_samples_leaf	30	2	30	8	24	3
RF	n_estimators	381	381	381	496	354	496
	criterion	gini	gini	gini	gini	entropy	gini
	max_depth	18	18	18	27	23	27
	min_samples_split	9	9	9	13	9	13
	min_samples_leaf	8	8	8	1	4	1
	max_features	auto	auto	auto	–	auto	–
XGB	eta	0.00718	0.00718	0.00718	0.00283	0.1966	0.000718
	n_estimators	868	868	868	905	107	913
	max_depth	5	5	5	20	19	27
	gamma	6.16×10^{-4}	6.16×10^{-4}	6.16×10^{-4}	6.11×10^{-5}	0.2567	0.2682
	subsample	0.510	0.510	0.510	0.524	0.765	0.776
	colsample_bytree	0.849	0.849	0.849	0.212	0.808	0.262
KNN	n_neighbors	4	42	4	4	11	2
	weights	distance	distance	distance	distance	distance	uniform
	p	1	3	1	5	1	5
LR	solver	newton-cg	lbfgs	liblinear	newton-cg	sag	saga
	penalty	–	–	l1	–	–	l1
	C	2.09×10^3	9.77×10^4	0.197	0.0715	0.6943	393.79

Table A.2: Final hyperparameters of the TabR deep learning model for the SNP and Gene P/A datasets. Hyperparameters were identical across both datasets, suggesting architectural robustness to different genomic feature representations.

Category	Hyperparameter	SNP Dataset	Gene P/A Dataset
Training	seed	0	0
	batch_size	256	256
	patience	16	16
	n_epochs	100	100
	context_size	96	96
Model	d_main	256	256
	d_multiplier	2.0	2.0
	encoder_n_blocks	0	0
	predictor_n_blocks	1	1
	context_dropout	0.2	0.2
	dropout0	0.2	0.2
	dropout1	0.0	0.0
	mixer_normalization	auto	auto
	normalization	LayerNorm	LayerNorm
	activation	ReLU	ReLU
	num_embeddings	–	–
Optimizer	type	AdamW	AdamW
	lr	0.0003	0.0003
	weight_decay	0.0001	0.0001